

Technology Stack

Date	27 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap	Responsive and user-friendly interface for farmers to enter soil and weather data.
2	Backend / Server-Side	Python, Flask	Processes user inputs, integrates the ML model, and generates crop recommendations.

3	Database / Data Storage	MySQL	Stores user information, prediction history, and agricultural datasets efficiently.
4	Cloud / Hosting / Deployment	Render / Railway	Easy deployment, scalability, and reliable hosting for the web application.
5	Version Control & CI/CD	Git, GitHub	Source code management, version tracking, and team collaboration.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, OpenWeather API (optional)	Machine learning model development, data preprocessing, visualization, and weather data integration.